- *Chemistry*: Computational chemistry, Process simulation
- *Climate science:* Weather modelling and climate simulations
- *Cybersecurity:* Code breaking (Shor's Algorithm), Cryptanalysis, Cryptography, Quantum Key Distribution (QKD), Quantum communication
- *Data science:* Big Data analytics, Search and optimization
- *Disaster management:* Earthquake and disaster prediction
- *Electronics:* Semiconductor design, Circuit simulation
- *Energy:* Energy optimization, Smart grid management
- *Finance:* Financial portfolio optimization, Finance fraud detection, Risk analysis and Monte Carlo simulations, Data sampling
- *Healthcare:* Drug discovery, Process reaction simulation, Genomics and precision medicine, Healthcare diagnostics, Protein folding analysis
- *Industry 4.0 and Industry 5.0:* Quantum digital twins and simulation
- *IoT:* Network optimization, Smart city simulation, Resource optimization
- *Logistics:* Supply chain management and optimization
- *Manufacturing:* Process optimization
- *Robotics:* Path planning, Resource optimization
- *Smart cities:* Urban planning optimization, Traffic flow management
- *Space:* Space communication, Satellite communication and optimization, Space mission planning
- *Telecommunications:* Telecommunication network optimization
- *Transportation:* Vehicle routing problems and resource allocation

Data science and engineering using quantum computing

Quantum computing and information processing now aids real-time data engineering and analytics — multi-dimensional data can be processed with accurate predictions (Table 2).

Table 2: A comparison of the quantum and classical methods of information processing and analytics

Analytics problem	Quantum approach	Classical method
Search	Grover's Algorithm	Linear Search
Optimization	QAOA	GA, ACO, PSO
Linear algebra	HHL Algorithm	Matrix Solvers
Clustering	Quantum K-Means	K-Means
Classification	Quantum SVM	SVM
PCA	Quantum PCA	PCA

It leverages the principles of entanglement, superposition, quantum gates and quantum parallelism. While classical computing systems process information using bits, quantum computers use qubits. In quantum information processing multiple states can exist simultaneously. This gives it the potential to navigate and accelerate high performance computational tasks, specifically for research that involves simulation, optimization and multi-dimensional data processing.

Quantum machine learning (QML)

Quantum machine learning (QML) integrates and makes use of hybrid models that combine machine learning techniques with quantum-based algorithms. Hybrid quantum models like Quantum Neural Networks (QNN) and Variational Quantum Classifiers (VQC) use high performance quantum algorithms for data preprocessing and optimization, and implement quantum circuits to execute machine learning and deep learning tasks.

Popular frameworks for quantum machine learning, quantum data engineering, and related tasks are:

- PennyLane, *https://pennylane.ai/*
- Qiskit, *https://www.ibm.com/quantum/qiskit*
- Qiskit Machine Learning, *https://qiskit-community.github.io/qiskit-machine-learning/*
- Tensorflow Quantum, *https://www.tensorflow.org/quantum*
- Google Cirq, *https://quantumai.google/cirq*
- OpenFermion, *https://quantumai.google/software*
- ProjectQ, *https://projectq.ch/*
- sQUlearn, *https://squlearn.github.io/*

Pennylane: Open source framework for quantum machine learning (QML) and quantum data engineering (QDE)

URL: https://pennylane.ai/

This open source framework is used for quantum data analytics and quantum information processing with a focus on quantum machine learning, quantum AI, and similar applications.

Pennylane can be used for real world AI based and intelligence applications including computer vision and pattern recognition, financial modelling, hybrid AI applications, natural language processing (NLP), optimization problems, quantum chemistry simulations, quantum data science, quantum machine learning, robotics, autonomous systems, etc.

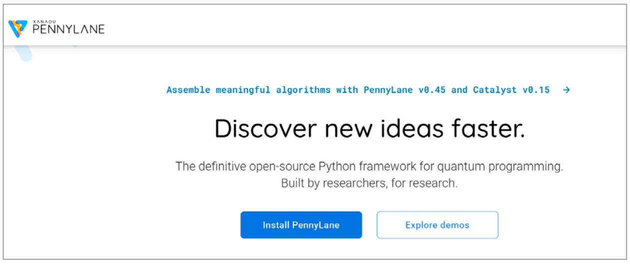

Figure 2: Pennylane — the open source framework for QML